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Review

Rethinking Blockchain Technologies for the Maritime Industry: An Overview of the Current Landscape

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Abstract: This survey aims to provide an up-to-date and succinct yet informative overview of the blockchain technologies for the maritime industry. We synthesize the recent advancements in blockchain development and its adoption across maritime sectors, highlighting the key blockchain use cases, including promoting maritime sustainability and optimizing maritime supply chain management through improved traceability, advancing smart shipping with automated processes and fostering collaboration among stakeholders by enhancing transparency. Through an analysis of current implementations, pilot projects, and case studies, we especially focus on identifying the challenges and barriers, reasoning on the status quo, and the opportunities and future perspectives for blockchain in maritime.

Keywords: blockchain; maritime; maritime digitalization; shipping industry; maritime safety; maritime operation efficiency; maritime decarbonization and sustainability

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1. Introduction

Trade has served as a crucial part of fueling economic growth and impacting human life, not only for the exchange of goods but also for the interchange of cultures and ideologies. Ninety percent of world trade is transported by sea and is estimated to be valued at USD 1.8 trillion annually [1]. Maritime trade involves many corporations in different countries, so vast amounts of documentation are generated in the process of trade. Despite the presence of national logistics information systems, it remains impossible for all participants to comprehensively monitor and share the entire process in real time. Reducing supply chain barriers to trade could potentially increase the global GDP by 4.7% (approximately USD 2.6 trillion) and global trade by 14.5% (approximately USD 1.6 trillion). This impact is projected to be more effective than the removal of tariff barriers, which is estimated to increase global GDP by 0.7% and global trade by 10.1% [2].

Blockchain is proposed as a disruptive technology that can solve the existing challenges in maritime trade. Most of the current maritime transportation processes are managed manually, which takes significant time and costs. Blockchain can enhance security, efficiency, and transparency in the maritime industry [3]. By applying blockchain to maritime logistics, shipping and transportation information can be distributed and permanently recorded within the blockchain. This allows all transaction participants to share and track logistical locations and status in real time. For example, for maritime trade from Kenya to the Netherlands, approximately 30 different trade participants are required to engage in over 200 transactions, with 10 days of document processing for shipment. However, in 2016, Maersk estimated that the utilization of blockchain could reduce total transportation

costs by 20%, saving USD 27 billion annually in the transportation process between East Africa and Europe. Additionally, the data stored on the blockchain cannot be modified, as it is managed transparently without tampering [4]. By using a blockchain, fraud issues prevalent in existing systems, such as information forgery or theft, can be greatly reduced [5].

Blockchain-enabled smart contracts automatically facilitate transactions upon meeting the predetermined conditions, simplifying and expediting traditionally complex transaction processes. In addition to facilitating transactions, blockchain's decentralized system ensures data is immutable and highly secure. When combined with other 4th Industrial Revolution technologies like IoT and AI, the future of the logistics industry will transform towards being more intelligent and sophisticated [6]. Many processes, from customs clearance and payment processing to cargo verification and validation, can be automated. AI technologies can be applied to forecast potential disruptions, optimize routes, and reduce delays, while real-time tracking data can be securely logged [7]. Blockchain combined with AI will reduce unnecessary time and costs, ensuring transparency and trust across the entire supply chain. Furthermore, the adoption of blockchain in maritime trade also contributes to environmental sustainability by reducing paperwork and incentivizing green processes, leading to lower CO2 emissions [8]. As blockchain technologies become more adopted, it could also lead to global standardization in logistics operations, making the industry more efficient and environmentally friendly [9,10].

Blockchain technologies have been successfully applied in a variety of industries. As illustrated in Figure 1, the leading sectors include finance, healthcare, government, retail, manufacturing, etc. Blockchain can revolutionize the financial sector by improving legacy workflows and transparency, reducing fraud and facilitating cross-border payments [11]. For healthcare services, blockchain technologies can improve patient data management, enable efficient clinical trials and drug traceability, and support personalized medicine [12,13] etc. Blockchain technologies can also enhance public services, facilitating identity and asset management and improving data security and procurement processes [14,15]. For retail and manufacturing, blockchain solutions can improve supply chain transparency, facilitate inventory management, enhance quality control and customer trust [16] etc. However, the prevalent adoption of blockchain still faces challenges, including the lack of standard development tools, diverse definitions and implementations, immature technology, resistance to change within corporations, specific enterprise needs, regulatory and legal hurdles, and unclear return on investments [17].

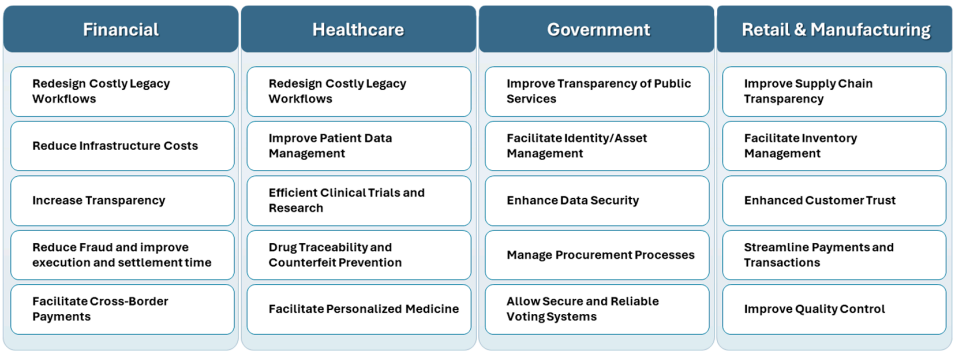


Figure 1. Excellent Application Verticals of Blockchain Technologies.

The current shipping processes are illustrated in Figure 2. The movement of goods from the exporter to the importer involves numerous procedures and intermediaries. It begins with the exporter and involves export freight forwarders managing logistics, documentation, transport booking, etc. The goods pass through export customs to ensure compliance with export regulations before moving via inland transport to the export port. Here, a carrier, typically a shipping or logistics company, transports the goods internationally to the import port. Upon arrival at the destination port, the goods clear import customs

where regulations and tariffs are assessed. Then, followed by import freight forwarders who facilitate the final logistics, such as inland transport to the importer. Services supporting entities in this process include insurance companies (to cover potential loss or damage), financial institutions (to handle payments and trade finance), and legal and regulatory authorities (to enforce trade regulations and compliance). This streamlined sequence of interactions highlights the collaborative effort required to move goods securely, legally, and efficiently across borders. However, it is worth mentioning that currently, this process is conducted mostly in a paper-based manner, which incurs significant costs and time. This is why many shipping companies are striving to adopt blockchain technologies in their trade operations to secure a leading position in the blockchain maritime platform market. By associating the application domains presented in Figure 1, it is obvious that shipping businesses can cover many application verticals that blockchain technologies have utilized for operational automation and efficiency, such as financial, insurance, government services, etc. Those blockchain solutions are applicable and promising for maritime shipping industries as well.

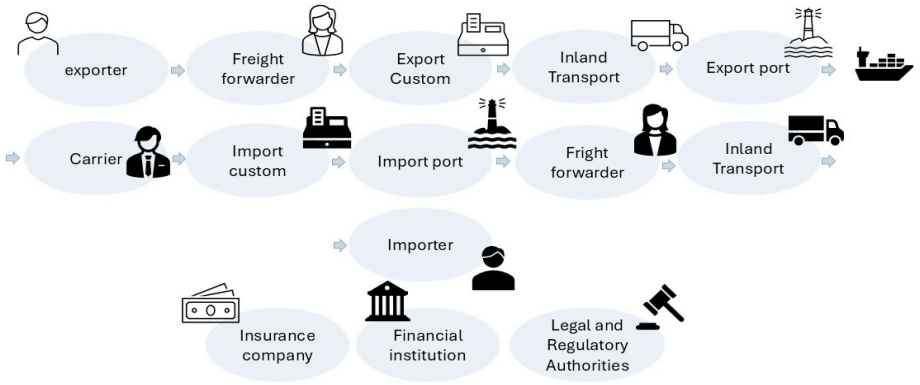


Figure 2. Current Processes in the Maritime Industry.

In this paper, we present a survey of the latest progress of blockchain solutions in the maritime sector, provide a scan of blockchain use cases in maritime, offer insights into the status quo and explore future directions of how blockchain can function in the maritime industry. For clarity, as illustrated in Figure 3, the contents of this paper are organized as follows: Sections 1 and 2 provide an overview of the background and motivation related to maritime digitalization and advanced information and communication technologies (ICT) technologies such as artificial intelligence, and blockchain technologies are demanded in this process. Section 3 explores specific blockchain applications in maritime, discussing the major projects of blockchain-based implementations for maritime applications. Section 4 analyzes the potential obstacles and benefits of blockchain in maritime, along with future directions. The survey concludes in Section 5 with a summary of the key analyses and insights.

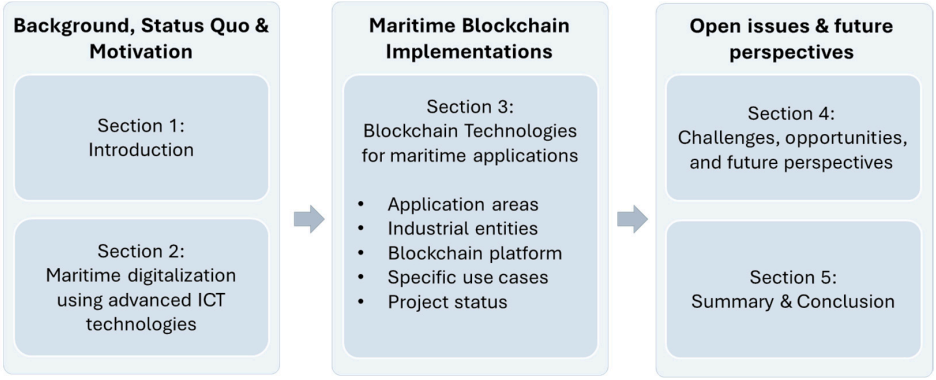


Figure 3. The Content Organization of the Study.

2. Maritime Digitalization

The maritime sector is undergoing a significant transformation and digitalization process with advanced Information and Communication Technologies (ICT) technologies [18–20]. This digitalization process is reshaping traditional maritime operations to be more efficient, transparent, and sustainable. With the development of technologies such as the Internet of Things (IoT), big data analytics, blockchain, and artificial intelligence (AI), the maritime sector is moving towards becoming more interconnected. These advancements enable the real-time monitoring of vessels, predictive maintenance, and optimized route planning, which reduce fuel consumption and operational costs, etc. Digital platforms facilitate better coordination among stakeholders, enhance supply chain visibility, and improve the overall safety and security of maritime operations, addressing the growing demands for efficiency and sustainability in global trade and logistics, ultimately positioning the sector to meet future challenges with greater agility and resilience [18–20].

It has been well-accepted that Artificial Intelligence (AI) and blockchain have become two of the most trending and disruptive technologies nowadays that may fundamentally reshape how the way we live, work and interact with the world [21]. The recent resurgence in AI is driven by breakthroughs in deep learning technologies and is fueled by the significant digital data available from the smart systems of various businesses (relying on sensing and IoT devices), social media, web applications [22], etc. AI-powered solutions are reshaping maritime operations in many areas, such as optimizing vessel scheduling, route planning, predictive maintenance, etc. AI-driven solutions help port authorities and shipping companies manage and predict traffic flow, reduce waiting times, and improve cargo handling efficiency. In the context of autonomous ships, AI is integral to achieving real-time situational awareness, risk detection, and collision avoidance, which are critical for ensuring vessel safety and compliance with regulations. These applications benefit from advancements in learning-based technologies, which leverage vast amounts of data from vessel movements, onboard equipment sensing, port operations and environment patterns.

Blockchain technology can manage a shared ledger of data, transactions, and logs in a decentralized, secure, and trusted manner, which can encourage collective data usage because it provides transparency, traceability and auditability regarding whose data is accessed, how the data are used, when, and by whom. Blockchain technology, rooted in the studies on cryptographically secured chains of blocks [23], was first applied and became popular for the digital currency network Bitcoin, launched in 2008 [24]. As the underlying decentralized data structure, blockchain is distinguished by its core characteristics of immutability, transparency and security, which ensures the integrity and reliability of the information stored on the blockchain, fostering trust in the system. In addition to its great success in the finance sector, the transformative potential of blockchain is evident across numerous sectors, where it promises enhanced traceability and audibility, operational efficiency, and security, especially in collaborative cross-organizational processes [25,26].

Blockchain can be categorized into public, private, and consortium networks. Public blockchains are fully decentralized and permissionless without a central authority, thereby allowing anyone to set up nodes and access the blockchain networks [27]. In the early stages of blockchain, a notable drawback of the public chain was its low transaction per second (TPS) performance, as it had to undergo a proof-of-work (PoW) consensus [28]. Nowadays, Layer 1 and Layer 2 scaling solutions [29] have been proposed for public chain networks to enhance the TPS performance. Layer 1 solutions include upgrades like modifying block sizes, altering consensus mechanisms like moving to proof-of-stake (PoS), or dividing the network into smaller sections for concurrent processing, i.e., sharding technologies. Layer 2 solutions [30] include rollups via bundling transactions or zero-knowledge proofs (ZKPs), sidechains, off-chain state channels, etc. In contrast, private blockchains are controlled by a central authority, permitting only authorized users to access the blockchain [31]. This centralized control facilitates faster processing speeds, easier network scalability, and the ability to assign different permissions to participants to protect data. Consortium blockchains [32] or federated blockchains represent a hybrid model

that combines features of both public and private blockchains. Consortium blockchains allow multiple organizations to maintain control over the blockchain network. While network participation is generally open, specific permissions can be granted to certain organizations [33].

The use of smart contracts is one of the critical building blocks for blockchain networks [34,35]. A smart contract is a decentralized, self-executing contract with the terms of the agreement directly written into code and run on a blockchain. The codes of a smart contract are stored on a blockchain and managed by network consensus for execution as programmed. In the context of smart contracts, Turing completeness means that a smart contract can execute any computational logic or process, allowing developers to create contracts capable of performing complex tasks and implementing intricate logic directly on the blockchain, which paves the way for the development and growth of Web3 applications [36,37].

The integration of AI and blockchain may eventually help create a collaborative ecosystem with advanced intelligence to enable better efficiency, automation, and security and enhance the quality of services in widespread application domains. Examples of the primary benefits of AI and blockchain synergy are summarized in Table 1. In the maritime sector, the integration of AI and blockchain holds transformative potential to address some of the industry's most pressing challenges and pain points. With the increasing digitization of global supply chains, ports, and shipping networks, the maritime sector has emerged as a prime candidate for leveraging these technologies to improve safety, efficiency, and sustainability. For instance, in port operations, AI can process historical data on vessel arrivals, congestion patterns, and weather conditions to optimize berthing schedules and predict potential delays. Blockchain ensures the reliability of the data sources and systems that produce the operational data, providing a trustworthy foundation for AI models. In global maritime supply chains, AI and blockchain can improve risk management by enabling real-time analysis of global events that may impact maritime operations, such as geopolitical events, cyber threats, port strikes and shipping events like blockage and attacks. This integrated approach enhances resilience across the maritime supply chain, empowering stakeholders to respond promptly to unforeseen events. The integration of AI and blockchain in the maritime sector not only promises increased traffic safety, operational efficiency and transparency but also aligns with broader sustainability goals. This synergy could drive the digital transformation of the maritime sector, positioning it to meet the demands of a complex, interconnected global trade ecosystem.

Table 1. Synergy of AI and Blockchain Technologies.

Blockchain for AI	Provides ideal storage for deep learning data and supports the maintenance of data integrity, provenance, reliability and security, leading to more trustworthy and credible AI decisions [38–40].
	Supports trustworthy data-sharing, reduces trust friction, and orchestrates AI solutions and decentralized intelligence across multiple parties [41–43].
	Enables secure and efficient knowledge management [44].
	Coordinates untrustworthy devices and edge-AI-enabled IoT [44].
	Improves trust in AI-powered robotic decisions and collective decision-making [45,46].
	Enables autonomous AI agents to engage and execute economic and financial transactions [47].
AI for blockchain	Makes blockchain applications more resilient against attacks [41,48,49].
	Supports blockchain network design and performance optimization, such as assisting in mining node pool decisions [50].
	Allows for better blockchain services with AI [51,52].

In Figure 4, a representative digital stack structured across several interconnected layers is illustrated for maritime digitalization. The infrastructure layer comprises IoT-

sensing hardware, information systems, and networks, which are responsible for gathering extensive data from various sources. These sources include Automatic Identification Systems (AIS) for ship tracking, coastal radar and Synthetic Aperture Radar (SAR) for maritime surveillance, meteorological and oceanographic data sensing systems, as well as data related to shipping, port operations, and human activities. This collected data is then fed into and managed with the data platform layer, which focuses on large-scale storage solutions tailored for spatial–temporal domain data. This layer also incorporates immutable storage, such as blockchain ledgers, to ensure data traceability and auditability, which are crucial for maintaining data integrity and compliance for cross-organization processes in the maritime ecosystem. Relying on the domain data, the model layer utilizes AI and advanced simulation and modeling techniques like Discrete Event Simulation (DES), Multi-Agent Systems (MAS), and Computational Fluid Dynamics (CFD) to create intelligent systems that support maritime operations. These models help predict, analyze, and optimize maritime operations.

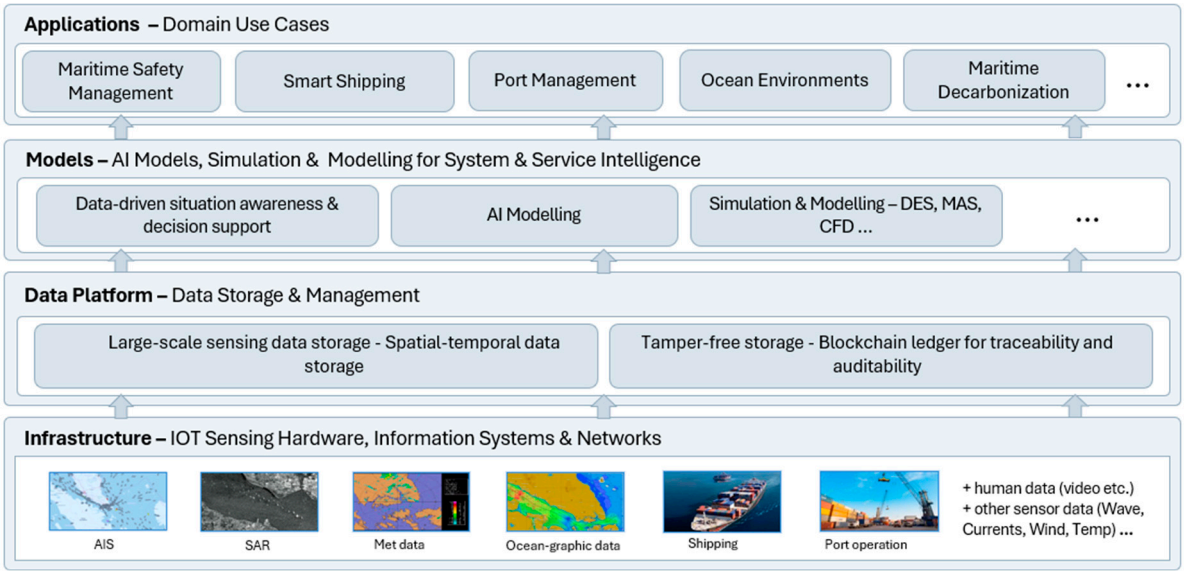


Figure 4. Advanced ICT Technologies for Maritime Digitalization.

Leveraging the data and model intelligence from the underlying layers, these technologies create a synergistic effect, support domain applications and use cases from maritime traffic safety management [53], smart shipping [54], port management [19], ocean environment monitoring, and decarbonization [8]. Collectively, these layers illustrate how data is collected, processed, and utilized to enhance various aspects of the maritime industry, integrating technologies like blockchain for secure data handling and AI for predictive analytics and services intelligence. Integrating and further substantiating the transformative potential of blockchain, AI, and IoT in the maritime sector are crucial for addressing the increasing complexity of maritime operations and meeting the demands of a globalized trade environment.

3. Blockchain Technologies for Maritime Applications/Use Cases

The features of blockchain technologies with respect to decentralization, immutability, traceability, transparency, and security can improve the existing operations in the maritime industry, while the primary applications can be broadly categorized into three areas: digitizing paperwork, enhancing information exchange, automating processes [55]. It is especially worth mentioning that the costs of paperwork were estimated at 15% to 20% of the total shipping fee [56].

By eliminating unnecessary processes and reducing excessive use of resources in maritime trading, consumers can receive goods more efficiently at lower costs. This also

leads to sustainable trade being a timely and important topic, contributing to slowing down global warming in the long term. The following discussion summarizes blockchain use cases into five categories: smart shipping, maritime sustainability/decarbonization, supply chain, ecosystem, and maritime finance. The diverse applications of blockchain in the maritime industry and current trends in blockchain will be highlighted.

3.1. Blockchain for Smart Shipping: CargoX

In 2023, approximately 2.2 billion tons of containers were traded by international seaborne, up from around 0.1 billion metric tons in 1980 [57]. Such extensive maritime trade is managed through the Bill of Lading (B/L), a crucial document in maritime trade transactions. The B/L serves as a contract for the transport of goods, proof of cargo receipt, and evidence of ownership, assuring all aspects of maritime trade agreements. It holds legal validity, can be used as collateral in Letter of Credit transactions, and acts as evidence in disputes concerning cargo transportation. However, traditional paper-based B/L handling faces delays in the issuance, delivery, and processing stages, leading to increased overall trade costs. The cost of this paperwork is estimated to be between 15 and 50 percent of the costs of physically moving the container around the world [56,58]. The overall time taken for processing documents was approximately 29% of the total delivery time from exporting farms to retailers [59]. Additionally, paper-based B/Ls are physically cumbersome to exchange, difficult to manage, susceptible to loss and damage, and prone to forgery.

To address these issues, companies are trying to transform BL into Digital B/L. The potential CO_2 reduction per electronic bill of lading (eBL) is approximately 27.9 kg, and about 16.9 kg approximately for an electronic Delivery Order (eDO) [60]. CargoX aims to resolve these problems by implementing Smart B/Ls using blockchain technology, offering a platform for importers and exporters to exchange B/Ls in a secure and reliable manner. CargoX envisions blockchain and cryptocurrency as pivotal in digitizing B/Ls, utilizing an Ethereum-based ERC20 encrypted permanent decentralized data repository. This open system facilitates the exchange of Smart B/Ls [61].

Despite ongoing efforts to adopt electronic B/Ls, issues like system reliability, electronic conversion, and program transparency have persisted. CargoX addresses these concerns through blockchain technologies, enhancing cost efficiency, transparency, reliability, and security. CargoX eBL can reduce many parts of the cost of maritime transactions. The cost of CargoX eBL was USD 15, which is almost 15% of the estimated usual price of the BL. Also, CargoX employs both off-chain and on-chain mechanisms to improve security and processing speed. All user data and public keys are stored off-chain, while original documents are kept for universal traceability on-chain. The CXO token comes with the CargoX platform, which is used for service fee payments processed via smart contracts. Of the utilized CXO tokens, 70% are burned, and 30% are returned to CargoX's treasury. CargoX incentivizes users by providing CXO tokens at 10 USD or 20% of the sales price for creating Smart B/Ls, thereby encouraging issuance. CXO tokens utilized by consumers are spent within the Ethereum network for Smart B/L issuance, with the remainder supporting operational costs or ecosystem development. This process facilitates smoother B/L issuance within the CargoX platform. By leveraging CXO tokens, CargoX enhances the efficiency of B/L issuance, aligning with their broader goal of revolutionizing the maritime trade document process and exchange through blockchain technology.

3.2. Blockchain for Maritime Sustainability/Decarbonization: BunkerTrace

The maritime industry pursues environmental sustainability through decarbonization. In 2019, the shipping industry and the financial sector enacted the Poseidon Principles to address greenhouse gas emissions. These principles require financial institutions to consider climate-related factors when making lending decisions for the shipping industry. Annually, the shipping industry reports its climate-adjusted status, which allows financial institutions to influence loan terms based on the environmental performance of their portfolios. Currently, 35 signatories are participating in the Poseidon Principles.

Additionally, companies are putting efforts into achieving decarbonization through technological innovations that reduce unnecessary energy consumption in maritime transport. Blockchain technology can contribute to decarbonization by enhancing trace, tracking, and transparency of sustainable fuel use within the maritime sector. Netherlands-based BunkerTrace uses synthetic DNA and blockchain technology to clarify the origin, composition, and quality of fuels. The company provides detailed information about fuel origins, quality, and tracking to ensure security and protection. By applying non-invasive synthetic DNA tagging to fuels and recording this information on the blockchain, BunkerTrace enables clients to continuously track fuel data, mitigating concerns about fuel adulteration. This tracking capability helps ensure the correct use of fuel, prevents the trading of low-quality illegal fuels, and promotes the transparency of fuel transactions, ultimately fostering the use of sustainable fuels [62].

BunkerTrace demonstrated the efficacy of its technology through a 2020 pilot project with Cargill, which tracked compliance with environmental regulations for Cargill's marine fuels. Since then, BunkerTrace has collaborated with various organizations, including S&P Global Platts, Singapore Shipping Association (SSA), and DNV GL, to enhance fuel transparency.

3.3. Blockchain for Maritime Supply Chain

3.3.1. TradeLens

In May 2017, Maersk and IBM undertook a project to verify the effectiveness of integrating blockchain systems into Maersk's container maritime transportation by combining Hyperledger Fabric, one of the consortiums/permissioned blockchain technologies, with IBM's proprietary technology. The purpose of TradeLens is to digitize paper-based processes in supply chain management and the maritime sector [63]. Figure 5 illustrates the blockchain-based business network developed by TradeLens. They aimed to create a blockchain-based solution that digitizes all contracts involved in maritime trade and builds a system capable of tracking the movement of containers dispersed worldwide.

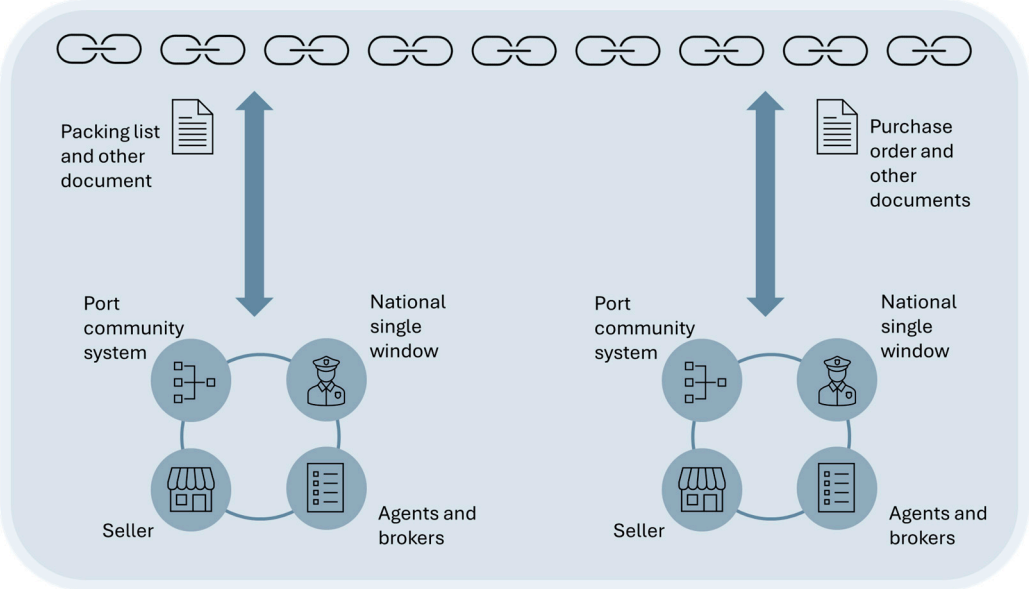


Figure 5. TradeLens Blockchain Business Network (re-created with inspiration from [64]).

However, in the first quarter of 2023, the TradeLens platform was ultimately discontinued. The project was initiated to advance the maritime sector through blockchain and succeeded in creating a viable platform, but it failed to achieve international industrial cooperation [65]. Consequently, TradeLens did not secure the commercial viability necessary to achieve its financial goals. The project, led by a major company like Maersk,

made many companies hesitant to participate in the industry. None of the Asian container shipping companies joined TradeLens, and Chinese shipping companies were dispersedly participating in the Global Shipping Business Network (GSBN). The failure to digitize documents also spans multiple jurisdictions, such as Electronic Bills of Lading.

Figure 6 illustrates the volume of documents and containers processed by TradeLens as of August 2024. Despite the setbacks mentioned, TradeLens holds its significance in that it applied blockchain to the traditionally conservative maritime industry, establishing a foundation and providing an example of the potential for further development.



Figure 6. TradeLens Achievement Numbers (re-created with numbers from [61]).

3.3.2. Walmart's Blockchain

Walmart also integrated blockchain technology into its supply chain, achieving significant success. In collaboration with IBM in 2017, Walmart developed a blockchain-based supply chain using Hyperledger to combat food contamination issues. For example, in the case of pork in China and mangoes in the Americas, Walmart's blockchain system reduced the time required to trace the origin of mangoes from seven days to just 2.2 s [66]. Furthermore, the transparency offered by the blockchain-enabled supply chain system enhanced consumer trust in the products.

Similar to the TradeLens project, Walmart implemented blockchain technology in its supply chains with IBM as a partner, but the outcomes were markedly different. Walmart adopted blockchain to address the specific challenge of food contamination, a problem that was difficult to resolve through traditional systems but for which blockchain provided an effective solution. Walmart established clear goals and demonstrated the utility of blockchain in supply chain management through rigorous testing. Additionally, Walmart encouraged its suppliers to adopt the blockchain system, and because the system was operated under Walmart's leadership, the adoption process was straightforward. Walmart's centralized management of the blockchain ensured consistent application of rules and procedures, leading to efficient operations, cost savings, and enhanced trustworthiness of the supply chain [67]. This approach enabled Walmart to realize direct value from the adoption of blockchain technology.

In contrast, TradeLens was developed to streamline data management in maritime trade. However, we faced significant technical, legal, and operational challenges to force participation from a diverse group of stakeholders (shipping companies, ports, customs, shippers, etc.) into a single platform. The goals of TradeLens were not clearly defined, and it failed to engage all relevant stakeholders. Although TradeLens was a project led by two major companies, Maersk and IBM, other significant shipping lines and ports either did not adopt the system or chose not to collaborate due to competitive concerns [68]. As a result, the platform's network effects were limited, reducing the system's overall efficiency. TradeLens' reliance on industry-wide cooperation, without enforcing such cooperation or establishing clear rules and procedures, led to conflicts among stakeholders. Ultimately, despite its ambition to develop a widely used platform for maritime trade, the project lacked a clear revenue model, which hindered stakeholder participation and led to the termination of the platform.

3.4. Blockchain for Other Applications in Maritime Ecosystem: Maritime Blockchain Labs (MBL)

The maritime shipping ecosystem is complex and varies significantly across different countries. Figure 7 shows the various elements of the maritime ecosystem. Due to the diverse interconnections among companies, nations, and environmental factors, defining a

single, uniform ecosystem is challenging. Each stakeholder within this ecosystem fulfills a distinct role, which is heavily influenced by surrounding factors. Furthermore, stakeholders often take on multiple roles depending on the circumstances. The key components of the maritime ecosystem include policymakers, regulators, ports, shippers and shipping agents, ship owners and operators, shipbuilders, fuel providers, researchers, etc. These entities all play crucial roles—each of them contributes to the overall functionality and resilience of the maritime industry [69].



Figure 7. Maritime Shipping Ecosystem (re-created with inspiration from [69]).

Maritime Blockchain Labs (MBL) [70] was established to address challenges within the maritime shipping ecosystem by integrating blockchain technology. MBL focuses on enhancing efficiency, regulatory compliance, and safety management in the maritime industry through the development and testing of blockchain-based solutions. MBL collaborates with a wide range of stakeholders, including ship owners, operators, and regulatory bodies, to create blockchain solutions that can be broadly implemented across the industry. Currently, MBL pilot projects are being conducted in three areas: the Marine Fuels Assurance Pilot, the Crew Certificate Management Pilot, and the Misdeclaration of Dangerous Goods Pilot.

The Marine Fuels Assurance Pilot focuses on ensuring the quality of marine fuel from the terminal to the vessel using blockchain technology. To achieve this, the MBL project developed a blockchain-based digital bunker delivery note specifically designed for the marine fuel supply chain. This tool allows stakeholders to enter and review details of fuel transactions and transfers between terminals, bunker barges, and receiving vessels. By enabling intuitive access to fuel-related information, the project enhanced the reliability of marine fuel quality. The Crew Certificate Management Pilot aims to create a blockchain platform that verifies the identity, training records, and certificate issuance, management, and clearance processes of all crew members. By collecting and reorganizing information according to the diverse requirements of various countries and companies, this platform offers a secure and trustworthy solution for certifying seafarers, significantly simplifying the previously complex verification process. The Misdeclaration of Dangerous Goods Pilot evaluates how blockchain technology can support the proper documentation and declaration of dangerous goods. This project standardized the inputs and outputs required

for the safe end-to-end delivery of hazardous materials, thereby building trust in the custody chain. By leveraging blockchain, the project enabled the immediate handling of quality assurance inspections and data extraction at each stage without the need for traditional documentation. This prototype ultimately improved the safety of hazardous material transportation, reducing the risk of maritime accidents [71].

However, MBL is facing challenges in collaboration with other consortium members, a persistent issue in every maritime blockchain project. Members of the blockchain consortium are often reluctant to share information on platform [70]. Despite these challenges, MBL remains significant in advancing the maritime ecosystem by offering a variety of services to the maritime industry through blockchain technology. MBL's innovative approach holds key importance in leading the efficiency and sustainability of the maritime industry.

3.5. Blockchain for Ports: Rotterdam Port

The first-generation blockchain can only record simple banking transactions. However, second-generation blockchains like Ethereum can run smart contracts and decentralized applications (dApps) and endlessly apply changes to the supply chain business. Blockchain enhances the efficiency of logistics transactions by allowing transparent and immediate tracking of the location and status of goods from origin to destination. Various aspects of the supply chain currently employ blockchain technology. Everledger certifies luxury goods like diamonds and wine by tracking their origins via blockchain. Walmart's food supply chain and UPS's logistics network chain also use blockchain.

The Port of Rotterdam, one of the world's largest ports, is a smart port that runs on multiple IT technologies. The Port of Rotterdam achieves real-time management with the Naviporta blockchain platform in the supply chain. The blockchain uses immutability to record items being transferred, but in a way, the quick moves of warehousing goods allow real-time tracking of status and location. In addition, processing times are greatly reduced by using digital copies in place of paper files, resulting in substantial cost savings.

Maritime transport between Korea and the Netherlands was used to test blockchain technology by the Port of Rotterdam in cooperation with ABN Amro Bank Samsung SDS. Thanks to Samsung SDS's supply chain technology platform, Nexledger, they were able to monitor the movement and status of cargo in real time. Moreover, tracking the location and status of smart containers makes container management run smoothly [72].

3.6. Blockchain for Maritime Insurwave

The maritime industry faces various challenges, such as unpredictable ocean conditions, accidents, and piracy. It is important to protect products and personnel from challenges through marine insurance [73]. Insurance plays a critical role in enhancing the reliability of trade, thus fostering the industry's growth. However, the current maritime insurance system is complex because of huge documentation processes and lengthy procedures, driven by the need to coordinate among multiple stakeholders [74]. This complexity not only increases insurance costs, which are ultimately passed on to consumers but also creates information asymmetries due to the selective sharing of documents. Additionally, the decentralized storage of insurance-related data complicates management and impedes real-time access.

Since maritime trade is one of the oldest forms of trade, most of its processes, practices, customs, and policy terms are well-established, making them easily translatable into blockchain [75]. By integrating blockchain technology into marine insurance, maritime industries can get advantages. Blockchain enhances transparency by allowing all stakeholders to access insurance details in real time [76]. The use of smart contracts automates insurance agreements, speeds up processes, and reduces paperwork and human intervention, thereby reducing costs [77]. Furthermore, by improving the safety of maritime transport, insurance can facilitate trade by reducing associated risks. Centralizing data on a blockchain enables real-time tracking and instant access when needed. The implementation of blockchain in the maritime industry demonstrates its effectiveness in managing sensitive

insurance adjustments, particularly through its capacity to modify premiums in real-time in response to dynamic conditions, such as a vessel entering a high-risk area. With transactions being immutably and transparently recorded on the blockchain ledger, disputes can be resolved with greater speed and efficiency [77]. In cases of cargo loss, total loss, or contamination of products and goods, such as pharmaceuticals and food, smart contracts can be programmed to automatically trigger payments once the required notifications, verifications, and supporting evidence have been recorded on the blockchain ledger [75].

Insurwave is a pioneering example of a blockchain application in maritime insurance. Founded in 2018 through an agreement between EY and Guardtime, Insurwave uses a private blockchain to securely transfer data between authorized parties [78]. This approach enhances data integrity, reduces fraud risk, and improves transaction transparency. By processing insurance policies through smart contracts, Insurwave enables quick actions such as policy renewals, premium adjustments, and claim payments while minimizing human involvement and operational costs [78]. Insurwave, the first blockchain-based maritime insurance platform, exemplifies how blockchain technology can manage intricate insurance systems in the maritime sector.

As shown in Table 2, numerous companies are striving to integrate blockchain technology into maritime trade. As a transformative technology, blockchain offers the potential to revolutionize the shipping industry. Various companies are leveraging distributed ledger technology (DLT) and smart contracts to enhance overall efficiency within the maritime sector. The primary motivation behind the adoption of blockchain in these industries is to address the shortcomings of traditional shipping systems. The focus is primarily on improving the efficiency of paperwork, accelerating information sharing, and enhancing the reliability of existing systems. However, following the cessation of operations of TradeLens, once the largest blockchain platform in the maritime industry, during the first quarter of 2023, the sector has seen a shift towards being driven by startups. Unlike TradeLens, which attempted to address all aspects of the shipping industry with blockchain, these new ventures tend to specialize and focus on specific areas. This specialization has the advantage of potentially increasing efficiency in targeted fields, but it also presents challenges. Companies may struggle to implement blockchain solutions, as they would need to apply different, often incompatible, blockchain programs to various sectors of the shipping industry. Currently, blockchain is being developed on multiple platforms, and the lack of interoperability among these platforms makes it challenging for companies to adopt blockchain for seamless operations across different aspects of their business.

Table 2. Using Blockchain Cases in the Maritime Sector.

Application Areas	Specific Use Cases	Companies	Platform Used
Smart Shipping	Electronic bill of landing	TradeLens PIL	Hyperledger Fabric
		CargoX	Ethereum
		Blue Water Shipping Louis Dreyfus	Ethereum Quorum
		Bolero American President Lines (APL)	Corda
Decarbonization	Fuel quality	Bunker-Trace	Ethereum
		BLOC (Blockchain for Low Carbon)	Hyperledger Fabric
	Carbon credits	ClimateTrade	Ethereum

Table 2. *Cont.*

Application Areas	Specific Use Cases	Companies	Platform Used
Supply Chain	Supply Chain	Blockshipping	Ethereum
		DexFreight	RSK
		Marine Transport International (MTI)	Hyperledger Fabric
		ShipChain	Ethereum
		VeChain	VeChainThor
		e-Marine	Hyperledger Fabric
		TradeLens Amazon	Hyperledger Fabric
		Alibaba	Hyperledger Fabric
Maritime ecosystem	Ecosystem	Maritime Blockchain Lab (MBL)	* Not mentioned
	Ship Builder	Samsung Heavy Industries	Hyperledger Fabric & Ethereum
		BP (British Petroleum) Shell	Hyperledger Fabric
	Smart Port	Port of Rotterdam Port of Los Angeles Port of Valencia	Hyperledger Fabric
		Port of Singapore	Ethereum
		Port of Antwerp	Hyperledger Fabric, Ethereum
Maritime finance	Underwriting	Insurwave	Corda
		Eurapco Unity	Hyperledger
	Claims	B3i Services AG	Corda
	Fraud Reduction	RiskStream Collaborative (from RiskBlock)	Corda
	Cross-Border Payment	Ripple Labs, Inc	Ripple
	Ship Financing	Shipowenrs.io	Ethereum
	Escrow Account	300cubits	Ethereum

* MBL does not mention what platform was used in their system.

4. Challenges, Opportunities, and Future Perspectives of Blockchain for Maritime

4.1. Difficulties in Technology

First introduced in 2008, blockchain remains in its early stages of technological development. Rapid advancements are occurring in various aspects, such as blockchain programs, algorithms, consensus methods, and applications, indicating that blockchain technology has not yet been definitively established. While blockchain is currently in its nascent stage and is being applied across all industries to drive innovation, it remains to be seen whether blockchain innovation is a definitive breakthrough.

Adoption Cost—To utilize blockchain, an enabling environment must be created. The problem is that the initial cost of implementing new blockchain-based solutions is high [3,79–82]. Since blockchain is a sophisticated technology, it requires a significant amount of time to learn. Consequently, hiring experts who thoroughly understand blockchain or educating existing employees on blockchain incur high costs [83]. Developing and maintaining programs with blockchain also takes lots more expenses than expected. Therefore, the initial costs of adopting blockchain technology are significant. Particularly, blockchain is more supporting position than other prominent technologies. Blockchain supports existing technologies to make them more convenient rather than being perceivable. Additionally, companies that adopt blockchain technology need to use a lot of energy to use blockchain systems [76]. As a result, management may not see the justification for investing heavily in adopting this technology, and the port industry, still uncertain about blockchain technology, may find it a considerable burden [84].

Data and Privacy Protection Concerns—One of the key characteristics of blockchain is its transparency and immutability; once transaction records are placed on the blockchain,

they are accessible to all participants involved in the transaction. The maritime industry handles a wide array of sensitive information, including detailed transaction information, cargo-related documents, and financial records. If companies adopt blockchain systems, protecting data privacy could become challenging [76,84]. For the port industry, adopting blockchain, with its potential cybersecurity risks such as DoS attacks and concerns over data security, is not an easy decision [80]. If personal information is leaked, it can undermine the crucial trust on which the port industry relies, making it cautious about adopting blockchain. Despite email or calling not being 100% secure, these methods are still used to share sensitive information such as bank account details or passport copies. Even though blockchain can prevent tampering risks, maritime stakeholders who are used to existing technologies do not see the necessity for change through digital transformation.

Scalability—When transactions are conducted by global stakeholders, a significant volume of transactions and computations will occur. Excessive computations can slow down speeds and create scalability issues, which may hinder the immediate and convenient information sharing that was originally intended [80,81,84–87].

Interoperability—Global stakeholders currently conduct shipping transactions within their systems and platforms due to the absence of a global logistics platform. This means they use different blockchain systems to exchange information [76,84,88]. If different blockchain systems are used between two stakeholders, document sharing between these entities becomes difficult.

Immutability—One of the important characteristics of Blockchain is immutability. Once a block is created in the Blockchain, the information recorded in that block cannot be altered. Immutability ensures transparency and trust between transactions, but it also presents the problem of non-reversibility in cases of user errors or criminal situations [80,84,89].

4.2. Challenges Arising Maritime Sector

Conservatism—Conservative culture in the Maritime industry is a barrier to adopting blockchain technology. The industry's long history means processes are already made and well conducted. Maritime industry executives adhere to maintaining the existing systems [85]. The industry is considered a late adopter of digital technologies and lacks digital innovation [83]. Family-owned businesses, hierarchical structures between members, and structural and conservative atmospheres in maritime companies make it difficult to introduce a new technology innovation in the maritime industry [90]. Decision-makers in the maritime industry are hesitant to adopt new technologies for innovation because innovation must bring confusion first.

Lack of Human Resources and Knowledge—Blockchain is a new and advanced technology. In less than 20 years since its inception, many significant aspects of blockchain technology continue to evolve. As a result, there are fewer skilled technicians compared to other technologies. Additionally, blockchain is challenging to learn. Companies find it burdensome either to hire skilled technicians at high salaries or invest heavily in training existing employees in blockchain technology [81,82,84,85].

Regulations Uncertainties—In the maritime industry, various stakeholders, such as cargo owners, ship owners, brokers, banks, and port authorities, are intricately connected. Each participant follows the processes and practices of their industries; it makes it extremely challenging to create a single digital platform encompassing all stakeholders [85]. Since blockchain is a new technology, there are no standardized regulations or laws in place. Blockchain stores diverse information in a decentralized ledger to ensure immutability, raising ambiguity about which country's laws should be followed for specific transactions [80,90,91]. When using blockchain smart contracts for cross-border transactions, complex problems related to international governing law or individual country laws may arise. Additionally, various issues associated with cryptocurrencies have led to national regulations being imposed. Before establishing an international blockchain platform, the

maritime industry must collaborate to set regulatory standards, then form the basis for blockchain platform program development [76,84].

Lack of Support from Stakeholders—Numerous stakeholders in the maritime sector hesitate to participate in a blockchain maritime platform [92]. The industry remains tied to traditional, paper-based processes, with many participants still accustomed to handling tasks with physical documents and unwilling to change these practices. Many participants are reluctant to share their information, collaborate, communicate, and coordinate on the blockchain platform because they do not want to share data with competitors, which could inadvertently support them [84]. Additionally, the perception of blockchain innovation varies among stakeholders. For direct participants in maritime transportation, such as importers, exporters, shippers, and carriers, blockchain adoption presents an opportunity to reduce intermediary costs. However, intermediaries in port transport, such as freight forwarders, customs brokers, and banks, may have a negative view of blockchain adoption.

Revenue Reduction in Various Industries—The adoption of blockchain in the port industry poses a threat to the revenue models of clearing institutions such as banks and intermediaries, potentially leading to decreased bank revenues and reduced cargo storage income at ports. Blockchain enables direct transactions between sellers and buyers without going through various intermediaries. Consequently, digital transformation through blockchain could replace or reduce the jobs of numerous intermediaries who earn money by acting as middlemen, resulting in significant unemployment. This becomes a factor that hinders the collaboration of stakeholders [84].

By reviewing the descriptions above, Table 3 summarizes the challenges from both the aspects of technology and the domain in a succinct form.

Table 3. Challenges for adopting blockchain in Maritime.

Main Reason	Challenges	Sources
Technology	Adoption Cost	[3,73,84]
	Data and Privacy Protection Concerns	[76,84]
	Scalability	[80,81,84–87]
	Interoperability	[76,84,88]
	Immutability	[80,84,89]
Maritime	Conservatism	[83,85,90]
	Lack of Human Resources and Knowledge	[81,82,84,85]
	Regulations uncertainties	[73,80,84,90]
	Lack of Support from Stakeholders	[84,92]
	Revenue Reduction in Various Industries	[84]

5. Conclusions

Blockchain technology is an emerging technology with its special characteristics: transparency, decentralization, immutability, traceability, and security. The inherent characteristics of blockchain technology have the potential to bring a new wave of innovation to the stagnating maritime industry. Blockchain can enhance efficiency and reduce costs in the maritime industry by digitizing traditional paper-based processes. When integrated with AI, blockchain can yield even more effective outcomes. This paper explores the role of blockchain in the maritime industry, presenting various examples of how blockchain can benefit the sector.

However, the application of blockchain in the maritime industry is not without its challenges. Many companies in the maritime sector already recognize the potential benefits of blockchain. However, they hesitate to actively develop the new blockchain management system or change the current systems due to the numerous barriers associated with its early-stage implementation. For instance, Maersk and IBM attempted to form a blockchain

consortium in the maritime sector in 2016 but failed due to difficulties in information sharing between parties. The failure of the TradeLens project has left many large companies hesitant to start blockchain projects. Consequently, only start-up companies participate in developing blockchain systems within the maritime industry, while big companies cautiously invest money and give limited information to support new initiatives.

Although numerous papers have explained the advantages of blockchain in the maritime sector and industry stakeholders acknowledge its potential, there remains uncertainty about its applicability to real maritime systems. Companies are unsure whether blockchain will lead to increased profits and efficiency instead of the current system. As a result, while many companies are willing to support blockchain projects financially and with information, they are reluctant to take further steps to participate.

The primary reason blockchain technology has not yet been widely adopted in the maritime industry is that the industry continues to function effectively without it [84]. For example, the financial sector, where blockchain is most effectively utilized, has embraced technology because the benefits outweigh the costs. Blockchain's distributed ledger system offers advantages such as reducing unnecessary processes and thus saving time, money, and energy. However, the burden of sharing sensitive data with all parties and the inability to immediately realize these benefits can deter companies from adopting blockchain.

To successfully integrate blockchain into the maritime industry, it is crucial to address the low participation rates among companies. If the benefits of information sharing outweigh the potential risks, the adoption of blockchain would be smoother. If blockchain is to thrive in the maritime sector, there must be compelling reasons for stakeholders to join blockchain consortiums. Future studies need to focus on the attractive and immediate benefits of participation in blockchain projects. Once companies realize the direct advantages of their involvement, such as efficiency gains, cost savings, and environmental benefits, more stakeholders will be inclined to join and share their resources and information. This collective participation will eventually enhance the blockchain platform's power in a positive way, driving the maritime industry's further development and innovation.

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